KARNAVATI UNIVERSITY

NAAC GRADE A+ ACCREDITED UNIVERSITY

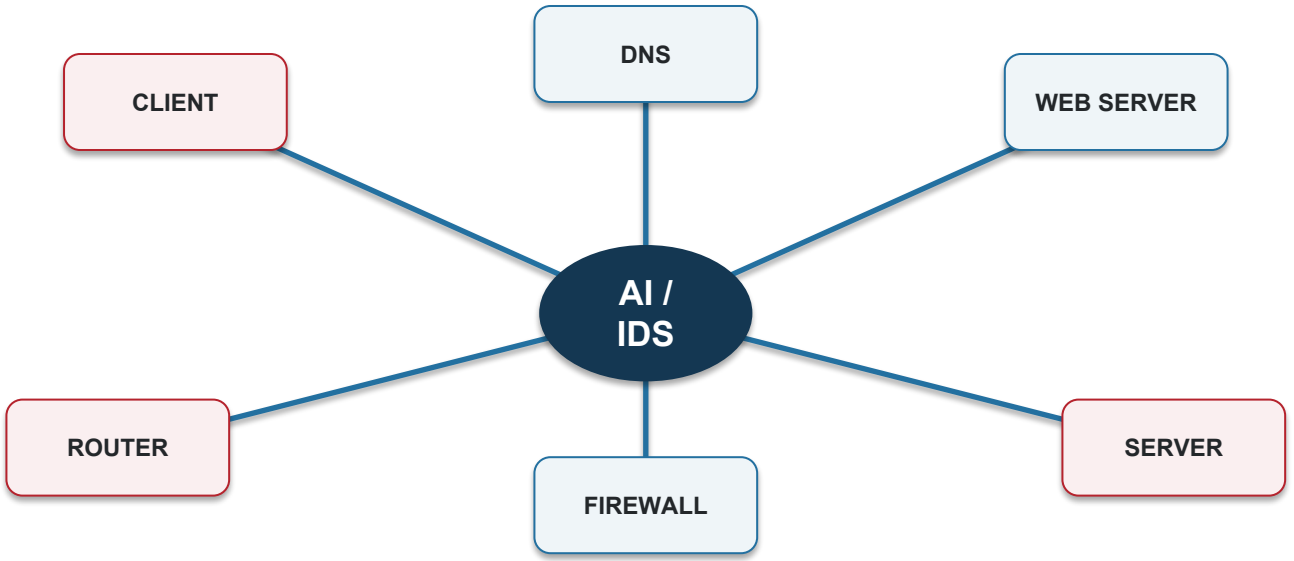
AI-Based Network Intrusion Detection System

Course: Computer Networks

Course Code: _____

Faculty: Ms. Pragya Vats

Group Members: _____



DISCUSSION & REVISION

- What is an Intrusion Detection System (IDS)?
- Why do we divide data into training and testing sets?
- What is the difference between precision and recall?
- What is a false positive in network intrusion detection?
- Which Computer Networks syllabus unit is most directly related to this project?

What is this project about?

Purpose

Use machine learning to identify network traffic that may be suspicious.

Networking link

Uses IP/flow information, protocols, packet counts and communication patterns.

Expected output

Predict Normal or Suspicious traffic and display the result.

Introduction to Network Security

Why do networks need monitoring?

Computer network

Devices exchange data using IP, TCP, UDP, DNS and other protocols.

Security challenge

Suspicious traffic can be difficult to identify manually in large networks.

AI opportunity

AI can learn patterns from examples and help analysts review traffic.

Problem Statement

The challenge we want to solve

Problem

Large volumes of traffic make manual inspection slow and difficult.

Need

A system that examines traffic features and flags flows that may be malicious.

Project scope

A learning prototype only; alerts need human review and testing.

KARNAVATI UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

UIT

Aim and Objectives

Project goals

Aim

Build an AI/ML prototype for network intrusion detection.

Objective 1

Prepare network-flow data and understand relevant features.

Objective 2

Train a classifier to distinguish normal from suspicious traffic.

Objective 3

Evaluate the model and visualize its performance.

Existing System and Limitations

Traditional monitoring approaches

Traditional approach

Known signatures, fixed rules and manual packet inspection.

Limitations

Rules need updates and unfamiliar attack patterns may be missed.

Balanced view

AI can also make errors and should support—not replace—security experts.

Proposed AI-Based System

Machine-learning workflow

Input

A labelled dataset containing network-flow features.

Preparation

Clean data, encode categories and create train/test splits.

Prediction

Train a model such as Random Forest to classify traffic.

Output

Show predictions, evaluation scores and flagged examples.

Computer Networks Concepts Used

Connection to the five syllabus units

Unit I

Application protocols and network services.

Unit II

TCP/UDP, ports, flow duration and packet counts.

Unit III

IP addressing, forwarding and source/destination information.

Unit IV

Link-layer communication and packet analysis.

Unit V

Intrusion detection, operational security and monitoring.

System Architecture

High-level design

1. Data

Use a public dataset or authorised lab traffic.

2. Features

Select relevant flow fields and prepare the data.

3. AI model

Train and test the classifier on separate data.

4. Dashboard

Display predictions, a confusion matrix and scores.

Dataset and Network Traffic Features

Use documented, appropriate data

Dataset option

CIC-IDS2017 is one possible public dataset; check its documentation before use.

Example features

Protocol, flow duration, packet totals, byte totals, packet lengths and flow rate.

Labels

Normal traffic and attack categories available in the selected dataset.

Data quality

Avoid leakage between training and test data.

AI/ML Algorithm and Methodology

Suggested beginner-friendly approach

Algorithm

Random Forest combines multiple decision trees.

Workflow

Load → clean → split → train → predict → evaluate.

Baseline

Compare against a simple model such as a Decision Tree.

Reproducibility

Record dataset version, features, split and model settings.

Tools and Technologies

Suggested implementation stack

Language

Python.

Libraries

Pandas, NumPy, Scikit-learn and Matplotlib.

Interface

Streamlit dashboard or a Jupyter Notebook report.

Network analysis

Wireshark can inspect authorised sample traffic; protect private data.

Implementation and Working

How the prototype will run

Step 1

Load data and check missing or invalid values.

Step 2

Prepare features and labels; encode categories if required.

Step 3

Split data into training and testing subsets.

Step 4

Train the model and predict labels for test data.

Step 5

Display sample predictions and performance metrics.

Results and Performance Metrics

Insert actual results after implementation

Accuracy

Overall proportion of correct predictions.

Precision

Of flows flagged as attacks, how many were attacks?

Recall

Of attack flows, how many did the model detect?

F1-score

A combined measure of precision and recall.

Results visual

Add your actual confusion matrix and measured scores.

Advantages, Limitations and Applications

What the project can and cannot do

Advantages

Automates first-pass review and supports repeatable testing.

Limitations

Results depend on dataset quality; false alarms and missed attacks can occur.

Applications

Security labs, learning environments and network administration support.

Ethics

Use authorised data and keep captured network information private.

KARNAVATI UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

UIT

Conclusion and Future Scope

Summary and improvements

Conclusion

The prototype demonstrates how ML can support traffic classification.

Future scope 1

Compare multiple models and attack categories.

Future scope 2

Add explainability to show which features influence predictions.

Future scope 3

Test on a separate dataset and evaluate false-positive rates.

KARNAVATI UNIVERSITY | NAAC GRADE A+ ACCREDITED UNIVERSITY

UIT

References

Suggested sources to cite in the final report

Dataset

Canadian Institute for Cybersecurity (CIC), CIC-IDS2017 dataset documentation.

Machine learning

Scikit-learn User Guide: Random Forest and model evaluation.

Packet analysis

Wireshark User's Guide and protocol documentation.

Final report

Add exact URLs, access dates, library versions and papers actually consulted.